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Machine Learning Field Notes

OCR Test Plates | October 2026



1. Closed-Form Ridge (L_2) vs. OLS Normal Equations

Given an augmented design matrix $\mathbf{X} \in \mathbf{R}^{n \times (p+1)}$ with leading intercept column of ones and target vector $\mathbf{y} \in \mathbf{R}^n$, Ordinary Least Squares minimizes residual sum of squares $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$. When predictors exhibit strong collinearity, the smallest eigenvalue ν_{min} of the Gram matrix $\mathbf{X}^T\mathbf{X}$ approaches zero, inflating parameter covariance $\text{Cov}(\boldsymbol{\beta}_{OLS}) = \sigma_\varepsilon^2(\mathbf{X}^T\mathbf{X})^{-1}$. Tikhonov Ridge ($L_2$) regularization adds a quadratic penalty $\lambda \Sigma \beta_j^2$ across the p slopes while leaving the intercept β_0 unpenalized ($(\mathbf{I}_0)_{00} = 0$):

$$\boldsymbol{\beta}_{\text{Ridge}}(\lambda) = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I}_0)^{-1}\,\mathbf{X}^T\mathbf{y}$$

2. Hoerl-Kennard (1970) Existence Theorem & Trace Variance

In the orthogonal eigenbasis of $\mathbf{X}^T\mathbf{X}$ with eigenvalues $\nu_1, \ldots, \nu_p$, the total trace variance and squared bias of the Ridge estimator are:

$$\text{tr}(\text{Cov}(\boldsymbol{\beta}_{\text{Ridge}})) = \sigma_\varepsilon^2 \Sigma_{j=1..p} [\, \nu_j / (\nu_j + \lambda)^2 \,], \quad \text{Bias}^2(\boldsymbol{\beta}_{\text{Ridge}}) = \Sigma_{j=1..p} [\, \lambda / (\nu_j + \lambda) \,]^2 (\beta_j^*)^2$$

Because $d(\text{Bias}^2)/d\lambda = 0$ at $\lambda = 0^+$ while $d(\text{Var})/d\lambda = -2\sigma_\varepsilon^2 \Sigma \nu_j^{-2} < 0$, there always exists $\lambda^* > 0$ that achieves strictly lower expected MSE than unregularized OLS.

3. Coordinate-Descent Lasso (L_1) Soft-Thresholding (Tibshirani, 1996)

Replacing the squared L_2 penalty with the L_1 norm $\lambda \Sigma |\beta_j|$ induces exact sparsity via the subgradient soft-thresholding update on partial residual correlation ρ_j:

$$\beta_j = S(\rho_j, \lambda) = \text{sign}(\rho_j) \cdot \max(0, |\rho_j| - \lambda)$$

Executed Industrial Turbine Benchmark Scorecard (*N* = 600, Seed 202610)

1. Point-in-Time Leakage Audit

On `industrial_turbine_telemetry.csv` (*N* = 600 hours across 30 units; chronological split n_{train} = 450, n_{test} = 150), the training-only median of `lube_pressure_bar` is **4.8700 bar** versus the leaky global median of 4.8300 bar. Including the post-event proxy `post_inspection_hours_LEAKY` fabricates a perfect Test ROC-AUC = 1.0000 and PR-AUC = 1.0000; removing it reveals the true sensor baseline.

2. Imbalanced ROC vs. PR Divergence

With only 7 faults vs. 143 normal hours in the test window (4.67% prevalence), the clean L_2 logistic classifier achieves ROC-AUC = 0.9630 because 5 False Positives divided by 143 negatives yields a tiny FPR = 0.0350. However, Precision-Recall AUC is **0.4939** (Precision = 5/10 = 0.5000 at τ = 0.30; Saito & Rehmsmeier, 2015). Lowering τ to the Bayesian cost optimum τ^* = \$5,000 / (\$5,000 + \$50,000) = **0.0909** catches 7/7 faults and reduces total test cost from \$350,000 to \$105,000.

Model / Pipeline	b_0	Load (b_2)	Vib (b_5)	Primary Metric	Secondary Metric	Status
OLS (λ=0)	27.1182	+7.2140	+2.1505	RMSE = 1.9235 °C	R^2 = 0.9530 (MAE 1.5578)	Executed
Ridge (λ=8.53)	27.1182	+6.9987	+2.2873	**RMSE = 1.9166 °C**	**R^2 = 0.9533 (MAE 1.5441)**	**Best Clean**
Lasso (λ=0.25)	27.1182	+6.7709	+2.4244	RMSE = 1.9327 °C	R^2 = 0.9526 (MAE 1.5608)	Sparse L_1
Clean Logistic	−3.8412	+1.1820	+1.4105	**PR-AUC = 0.4939**	ROC-AUC = 0.9630 (F_1 0.5882)	Clean Sensor
Leaky Logistic	−6.1204	+0.1102	+0.0940	PR-AUC = 1.0000	ROC-AUC = 1.0000 (Brier 0.0005)	Target Leak